

ANTICESTODAL PRINCIPLES OF *NIGELLA SATIVA*, LINN. (KALONJI) SEEDS

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ABSTRACT

Efficacy of the total glycosides, saponins and anthraquinones isolated from *Nigella sativa* (Kalonji) seeds was judged in the goats infected naturally with mixed gastro-intestinal nematodes. The glycosides were administered orally to groups of sheep in 100, 150 and 200 mg/kg body weight doses. Nilzan^(R) 15 ml/15 kg was similarly administered to the control sheep. Pre- and post-treatment faecal eggs per gram (EPG) counts and their percentage reductions were determined on 3rd, 10th and 15th day of drug administration. Their 100, 150 and 200 mg/kg oral dose of the glycosides reduced the EPG counts highly significantly ($P < 0.001$) on day 10 and 15. Statistically 150 and 200 mg/kg dose of *N. sativa* glycosides were as effective as Nilzan^(R) against goat cestodes on post-treatment days 10 and 15. However, a lower dose of 100 mg/kg of the glycosides could also produce results similar to Nilzan^(R) on day 15 only. The saponins and anthraquinones isolated from the *N. sativa* seeds, however, failed to exert any significant anticestodal effect in their 100, 200 and 300 mg/kg doses. These data suggest that the total glycosides of *N. sativa* seeds possess considerable activity against cestodes in the goats and may be responsible for the anticestodal efficacy of the powdered seeds already reported to be effective against tape worms of sheep.

INTRODUCTION

Modern anticestodal drugs have got neither complete curative properties nor they retain their efficacy on their continuous use due to development of tolerance (Waller and Prichard, 1985). In addition, they adversely affect milk and meat production of animals during course of their treatment and even long after their use (Blood *et al.*, 1983). Through ages man has looked towards the plants for treatment of diseases and even today they are being employed for the development of new drugs. Indeed, Pakistan is rich in herbal wealth and search for alternate anthelmintics from plants has now increased in importance as control of the parasitic infections involves the regular use of appropriate drugs (Michel *et al.*, 1981).

The seeds of *Nigella sativa*, Linn (Family, Ranunculaceae), commonly called

"Kalonji" have been used in "Unani" medicine and "Tibb-e-Nabvi" for the treatment of various diseases of man and animals (Said, 1969; Ghaznavi, 1988). The seeds have also been prescribed to treat various helminthic infections (Nadkarni, 1954; Awan, 1991; Satyavati *et al.*, 1987). Other medicinal properties ascribed to this indigenous plant in traditional medicine include stimulant, carminative, diuretic, emmenagogue and galactagogue. In addition, the crude drug has been used in puerperal fever, skin eruptions and scorpion-stings (Chopra *et al.*, 1956). Recent researches have proved several other medicinal properties of the seeds. Akhtar and Javed (1991) have reported that 2.5 g/kg dose of the powdered seeds, the equivalent amount of their ethanol extract and niclosamide (0.01 g/kg) were equi-effective for the treatment of sheep suffering from natural *Moniesia expansa* infection in sheep. Furthermore, the powdered seeds and their ethanolic extract were highly potent in treating cestodes in children (Akhtar and Riffat, 1991). However, active principle(s) responsible for this activity was not determined so far. Therefore, this study was conducted in goats to screen anticestodal efficacy of glycosides, saponins and anthraquinones isolated from the *Nigella sativa* seeds.

MATERIALS AND METHODS

Chemical isolation of glycosides, saponins and anthraquinones of *N. sativa* seeds

The best quality dried seeds of *Nigella sativa*, Linn. (Family: Ranunculaceae) sold in the market under the name "Kalonji" were purchased from a local herbal dealer and made then completely free from dust and other adulterations. The plant drug was kept in an incubator at 40°C for about 24 hours and was then finely powdered with an electric Irinder. Total Ilycosides of powdered *N. sativa* (Kalonji) seeds were isolated by the method described in detail by brain and Turner (1975). They were stored in air tight glass bottles at 4°C.

Animals used and testing procedure

Privately owned local mixed-breed flocks of beetal goats, located in and around the city of Faisalabad (Punjab) were used for the trial. Faecal samples of 250 goats which were found to have been kept under poor managerial and unhygienic conditions were collected per rectum by finger or at the time of defaecation. These faecal samples were screened to detect the cases suffering from gastrointestinal cestodes by using the direct smear method described by Soulsby (1982). In the first trial, randomly selected positive cases harboring amixedcestode infection in their gastrointestinal tracts were selected to evaluate the anticestodal efficacies of three different doses of the *N. sativa* glycosides. Then, anticestodal activity of the saponins and anthraquinous of *N. sativa* was also judged by using separate groups of goats taken from a different flock. All goats were randomly divided into groups of six animals each. The positive faecal samples were further examined for determining the pretreatment faecal EPG

(eggs per gram) counts by the McMaster egg counting technique after adding 30 ml of saturated salt solution as described in detail by Soulbsy (1982) and Stoll and Huscheer (1926). The EPG counting was carried out after about 8-12 hours of the sampling so as to get the eggs released naturally from the segments of *Moniezia* (tapeworms).

In all studies, a group of six goats was kept untreated as control while a second group was treated with a control drug, Nilzan^(R) (Levamisole HCl, 1.5% + Oxytoclozanide, 3% w/v) at a dosage level of 5 ml/15kg body weight orally. The remaining test groups were treated orally with three increasing doses of the *N. sativa* isolates. All test chemical entities were administered orally to the goats with a drenching bottle after suspending them in 1% gum tragacanth solution in water. After the drug administration, the post-treatment EPG counts of the faecal samples were again determined on 3rd, 10th and 15th days. Data were expressed as means $\pm$ SEM and the percentage reductions of the EPG counts were calculated. Student's 't' test was used to check the significance of the data according to Snedecor and Cochran (1980).

RESULTS

Total glycosides of Nigella sativa (seeds)

Three doses i.e. 100 mg, 150 mg and 200 mg/kg body weight of the glycosides isolated from *N. sativa* were tested. All the doses produced significant ($P < 0.05$) reduction in EPG counts on days 10th and 15th. The percentage EPG reductions were non-significantly ($P > 0.05$) different from NILZAN^(R) treated group on post-treatment days 3rd, 10th and 15th at dose rate of 200mg/kg. Non-significant ($P > 0.05$) difference from NILZAN^(R) was also noted on 10th and 15th days with 150 mg/kg body weight while similar result was obtained on only 15th day after treatment with 100 mg/kg of the glycosides (Table 1).

Total Saponins of Nigella sativa (seeds)

Three doses of the saponins i.e. 100 mg, 200 mg and 300 mg/kg body weight were administered to the goats. None of these dosage levels of total saponins isolated from *N. sativa* seeds showed significant ($P > 0.05$) reduction in EPG counts on 3rd, 10th and 15th days after the drug administration (Table 1).

Total Anthraquinones of Nigella sativa (seeds)

The anthraquinones were also administered orally the doses of 100 mg, 200mg and 300 mg/kg body weight and EPG counts were found to be non-significantly ($P > 0.05$) reduced on all the days checked (Table 1).

DISCUSSION

Helminth parasitic infections have continued to prevail in Pakistan and other

Asian countries. No doubt considerable progress has been made to control such infections in animals with the introduction of synthetic compounds (Booth and McDonald, 1983). However, anticestodal anthelmintics so far available for the treatment of such infections do not seem to possess ideal properties. Instead, they are well known to possess several adverse/side effects. Furthermore, the development of resistance to modern anthelmintics in animals against nematodes and cestodes has also been reported by several workers including Donald (1983), Waller and Prichard (1985), etc. The phenomenon has also strongly elucidated the need for exploring alternative therapeutic measures for the cestode infections in animals.

Many plant drugs have been used in this part of the world from ancient times for the treatment of parasitic infections in man and animals (Nadkarni, 1954; Chopra *et al.*, 1956; Said, 1969). Pharmacological experimentation on various indigenous medicinal plants allegedly used to treat parasitic diseases have already resulted in the discovery of several potent antiparasitic plant drugs (Akhtar, 1988; Akhtar and Javed, 1991; Akhtar and Ahmad, 1992). Some of these plants have been, therefore, phytochemically screened for the insulation of active principles. *Nigella sativa*, Linn. seeds which were previously shown to possess significant anticestodal activity (Akhtar and Javed, 1983; Akhtar and Riffat, 1991) have now been qualitatively and quantitatively screened for their phytochemical constituents. These substance has been tested for their activity against the cestode infections.

The data obtained have revealed that the three (3) substances including glycosides, saponins and anthraquinones are contained in the *Nigella sativa* seeds. However, only the glycosides of *N. sativa* could exhibit significant anthelmintic activity when administered orally to the goats suffering from natural gastrointestinal cestode infection (Table 1). In 150 and 200 mg/kg oral doses they have been found to be equi effective to Nilzan^(R) (Levamisole 1.5% and Oxytoclozanide 3%), a commonly used synthetic anticestodal drug on days 10 and 15 post-treatment. The saponins and anthraquinones isolated from *N. sativa*, however, failed to exert any significant effect on the cestodal infection when administered orally to the naturally affected goats. This has clearly showed that active principles of *N. sativa* seeds would be their glycosides and not their saponins and anthraquinones. The later, however, could have contributed towards the anthelmintic efficacy of the crude drug in some way. The separated saponins alone produced quite a high incidence of side effects in the treated goats (Table 2). From these results it may be assumed that the adverse effects of the saponins of *N. sativa* are negated in some way by some other constituents of plant drug including the glycoside making the whole powder and the glycosides more safe and free of side effects than the isolated saponins. Since the crude drug powder of *N. sativa* contained 0.04% glycosides, a much higher than the calculated dose of the isolated glycosides has produced a response non-significantly different from that of Nil-

Table 1
Mean $\pm$ SEM faecal eggs per gram counts and their %age reductions in goats suffering from Cestodal infection treated with glycosides, saponins and anthraquinones of *Nigella sativa* seeds (No. of goats in each group = 6)

Treatment	Dosage	Pre-treatment EPG counts (Zero day)	Post-treatment EPG counts			%age EPG reductions		
			3rd day	10th day	15 day	3rd day	10th day	15th day
Untreated	-	2152 $\pm$ 204	2142 $\pm$ 388	2152 $\pm$ 468	2332 $\pm$ 356	-	-	-
Nilzan ^(R)	5 mg/kg b.wt.	1486 $\pm$ 196	744 $\pm$ 108*	276 $\pm$ 60	36 $\pm$ 6**	51 $\pm$ 6	80 $\pm$ 5	97 $\pm$ 4
Glycosides	100 mg/kg b.wt.	1476 $\pm$ 216	1140 $\pm$ 172	532 $\pm$ 108*	264 $\pm$ 108	23 $\pm$ 4	64 $\pm$ 8	82 $\pm$ 7@
of <i>Nigella sativa</i>								
- do -	150 mg/kg b.wt.	1788 $\pm$ 364	1346 $\pm$ 180	516 $\pm$ 204*	312 $\pm$ 96*	30 $\pm$ 5	71 $\pm$ 8@	86 $\pm$ 9@
- do -	200 mg/kg b.wt.	1440 $\pm$ 252	984 $\pm$ 264	280 $\pm$ 46*	88 $\pm$ 24*	40 $\pm$ 7@	84 $\pm$ 5@	94 $\pm$ 5@
Saponins of	100 mg/kg b.wt.	1968 $\pm$ 344	1976 $\pm$ 374	1952 $\pm$ 252	1936 $\pm$ 120	0 $\pm$ 0	0.8 $\pm$ 0.4	1 $\pm$ 0.2
<i>Nigella sativa</i>								
- do -	200 mg/kg b.wt.	1776 $\pm$ 310	1696 $\pm$ 256	1966 $\pm$ 120	1633 $\pm$ 96	0.4 $\pm$ 1	1 $\pm$ 0.2	8 $\pm$ 4
- do -	300 mg/kg b.wt.	2316 $\pm$ 456	2270 $\pm$ 304	2292 $\pm$ 156	2212 $\pm$ 196	2 $\pm$ 1	1 $\pm$ 0.4	4 $\pm$ 2
Anthraquinones	100 mg/kg b.wt.	420 $\pm$ 84	432 $\pm$ 72	406 $\pm$ 48	411 $\pm$ 36	0 $\pm$ 0	4 $\pm$ 2	2 $\pm$ 1
of <i>Nigella sativa</i>								
- do -	200 mg/kg b.wt.	564 $\pm$ 160	588 $\pm$ 186	554 $\pm$ 96	530 $\pm$ 48	0 $\pm$ 0	2 $\pm$ 1	6 $\pm$ 3
- do -	300 mg/kg b.wt.	516 $\pm$ 180	500 $\pm$ 144	480 $\pm$ 76	490 $\pm$ 67	3 $\pm$ 1	7 $\pm$ 1	4 $\pm$ 2

* Significantly ($P > 0.05$) less than that of pre-treatment value

** Highly significantly ($P < 0.001$) less than that of pre-treatment value

@ Non-significantly ($P > 0.05$) different than that of respective value of Nilzan^(R) treated group

zan^(R). Thus it is evident that the *N. sativa* powder contains some synergetic compounds that enhance the efficacy of the crude drug. This seems to support the contention that plant drugs in natural form are more efficacious and safer than their isolated chemical entities. The present studies do also encourage the possible use of this cheap indigenous plant drug for the routine deworming and treatment of cestode infections in animals, especially the goats. Further chemical studies followed by pharmacological and toxicological investigations are yet needed to isolate the pure active principle(s) of *Nigella sativa* (Kalonji) seeds and to elucidate their exact mechanism of anticestodal action.

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